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Immersive and interactive cyber-physical system (I^2CPS) and virtual reality interface for human involved robotic manufacturing

Huitaek Yun [a], Martin B.G. Jun [a,b,*]

[a] Indiana Next Generation Manufacturing Competitiveness Center (IN-MaC), Purdue University, 1105 Challenger Avenue, Suite 400, West Lafayette, IN, 47906, USA
[b] School of Mechanical Engineering, Purdue University, 585 Purdue Mall, West Lafayette, IN, 47907, USA

A R T I C L E I N F O

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A B S T R A C T

Smart manufacturing promotes the demand of new interfaces for communication with autonomies such as big data analysis, digital twin, and self-decisive control. Collaboration between human and the autonomy becomes imperative factor for improving productivity. However, current human-machine interfaces (HMI) such as 2D screens or panels require human knowledge of process and long-term experience to operate, which is not intuitive for beginning workers or is designed to work with the autonomy. This study proposes a human interface framework of cyber-physical system (CPS) based on virtual reality, named as immersive and interactive CPS (I^2CPS), to create an interface for human-machine-autonomy collaboration. By combination of data-to-information protocol and middleware, MTConnect and Robot Operating System (ROS), heterogeneous physical systems were integrated with virtual assets such as digital models, digital shadows, and virtual traces of human works in the virtual reality (VR) based interface. All the physical and virtual assets were integrated in the interface that human, autonomy, and physical system can collaborate. Applying constraints in the VR interface and deploying virtual human works to industrial robots were demonstrated to verify the effectiveness of the I^2CPS framework, showing collaboration between human and autonomy: augmentation of human skills by autonomy and virtual robot teaching to generate automatic robot programs.

1. Introduction

The 4th industrial revolution has changed past manufacturing technologies which have focused on mass production and cost savings [1]. The concept of the revolution, smart manufacturing, emphasizes on intelligent autonomy with minimum human intervention [2,3]. Fig. 1 categorizes both previous well-known and recent smart manufacturing technologies according to the view of human, autonomy, and physical systems. Before the concept of the industrial 4.0, manufacturing execution system (MES) and supervisory control and data acquisition (SCADA) are widely used to monitor and control manufacturing processes. Enterprise resource planning (ERP) and Product lifecycle management (PLM) maintain the information of products and businesses. Although the technologies enable communication between the entities, creating more seamless collaborative environment among the entities are required for smart manufacturing. For example, intuitive decision of humans and big data analysis of autonomy can work together to improve productivity. Humans have cognitive and associative skills to analyze the circumstance instantly. However, technologies for autonomy such as artificial intelligence and machine learning are more advantageous to analyze the big data and construct prediction models than humans.

In current human interfaces, workers have been interacting with 2D screens for machines, teaching pendants for industrial robots, operation panels, and personal computers while monitoring and adjusting processes. The interaction between human and machine happens when workers press buttons, switches, and clicking of screens after their cognitive activity. The skilled interaction requires repeated training and long-time experiences to fully understand the operation of machines and processes. Recent studies have proposed intuitive methods for the issues using smartphones, tablets, Virtual Reality (VR) and Augmented Reality (AR) gears, voices, and gestures [3]. However, autonomy in smart manufacturing such as digital twin and data analysis is also separated from the interface. Therefore, there is a demand of new human interface for intuitive visualization, control, and collaboration with autonomy.

Fig. 2 illustrates the concept of the new interface. It is possible that human and autonomy can both understand real-time data and control

* Corresponding author at: Indiana Next Generation Manufacturing Competitiveness Center (IN-MaC), Purdue University, 1105 Challenger Avenue, Suite 400, West Lafayette, IN, 47906, USA.

E-mail address: mbgjun@purdue.edu (M.B.G. Jun).

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the system autonomously. However, the automation at both entities stems from the cognitive and associative phase of information, meaning that human and autonomy share the works from the same information or divide the information to separate the works. Another thing is that HMI relies on the cognition function of humans. On the other hand, the new interface is expected to help humans to enhance cognition and association of the information provided by digital shadow and data analytics.

This paper proposes the concept of immersive and interactive CPS (I^2CPS) framework for human-machine-autonomy interaction. The I^2CPS framework is the combination of Machine to Machine (M2M) middleware and data-to-information protocol, covering wide range of information from raw data to human-context. The goal of I^2CPS is to provide immersive interface to human using VR, which is the combination of the result of autonomy work and the control of autonomy, thereby human and autonomy can collaborate.

Chapter 2 reviews previous studies about smart manufacturing architecture, CPS, digital twin, data communication networks, human interfaces using virtual reality, and similar research areas. Chapter 3 proposes the I^2CPS framework using MTConnect and ROS middleware to construct a network between human with HMI interface, digital shadows, digital models, and multiple physical entities. Chapter 4 presents the developed applications using the platform to verify the effectiveness of the platform. Applying constraints to human works in virtual space improves the work accuracy, augments human cognition, and avoids human mistakes. In addition, the virtual works were converted to autonomous robot programs showing how humans teach autonomy. Chapter 5 discussed limitations and possible research opportunities of this study.

2. Literature review

2.1. Smart manufacturing architecture

Several studies have proposed the architecture of smart manufacturing. Lee et al. [4] introduced a 5-level CPS architecture. At the smart connection level, collecting data with different types and choosing proper sensors are important. The data-to-information conversion level infers meaningful information such as machine health and remaining tool life of CNC machines. Cyber level analyzes the collected information to compare and predict machine behavior. At the cognition level, the knowledge of system is accumulated and visualized to human using proper graphics. At last, the configuration level provides feedback to the system as corrective and preventive manners. Lu et al. [5] mentioned the vision of smart manufacturing with a brief architecture. Physical layer includes all the physical assets such as machines, robots, and sensors. The physical assets are connected altogether in the cyberspace gateway layer. Next, cyberspace layer extracts data for decision-making and optimization, modeling, simulation, data exchange, and data integration. The converted information is utilized to various applications at business and consumer sides. Similarly, Xu et al. [6] proposed Machine Tool 4.0 framework for smart machine tools. The objective of the framework to change tool path and machining parameters on the fly and to provide tools to measure machine performance and its degradation. Such functions are implemented by artificial intelligence in cyberspace. Kusiak [7] also mentioned the framework of smart manufacturing as 1) physical systems: manufacturing equipment and its local intelligence, 2) Interface to collect data and send decisions in a standardized way, 3) Cyberspace to provide system intelligence.

These smart manufacturing frameworks have similarities. First, physical assets are connected in smart ways, tether-free and plug-play. To collect data with heterogeneous types, various middleware technology and data standardization protocols are used. Second, cyberspace converts the collected data to information that the analysis, digital twin modeling, simulation, and decision-making are performed. It can be classified as 1) data mining and digital twin generation, 2) cognition, and 3) self-configuration. Third, applications are tools for humans such as factory worker, manager, entrepreneur, and consumer. Each application can access any information in cyberspace for its needs. In addition, visualization of the information is required to maximize human's ability [4].

The I^2CPS framework is based on the similarities of smart manufacturing architectures while emphasizing on collaboration between human and autonomy. In general, autonomy modules are inside the cyberspace, and human utilizes them from applications. Especially

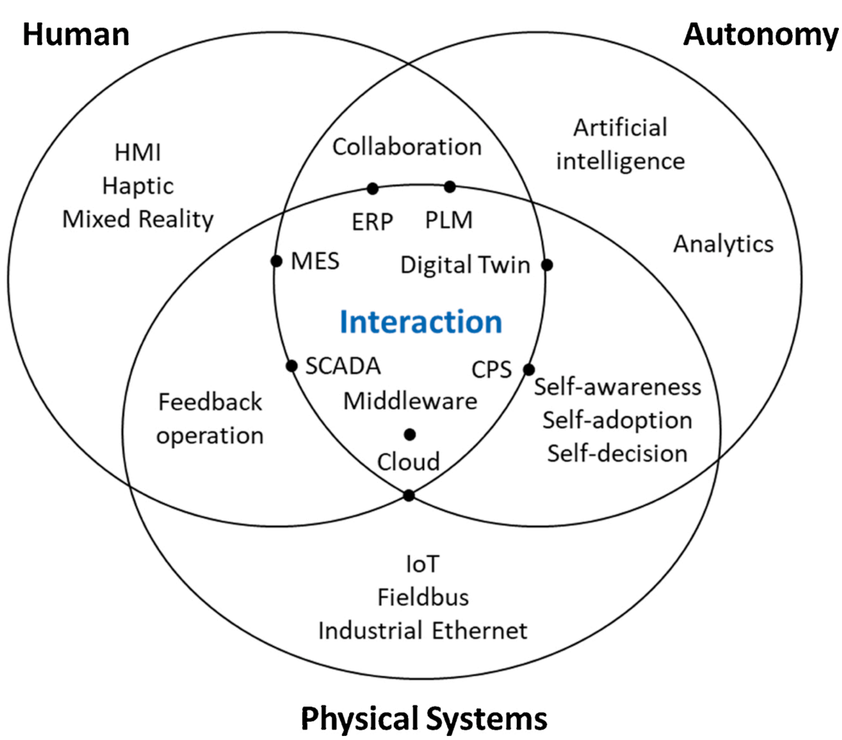


Fig. 1. Categorization of technologies for smart manufacturing.

in the I^{2}CPS framework, cognition of humans can be substituted to artificial intelligence, the autonomy can control manufacturing systems instead of human, and both human and autonomy share the work of data analysis and association.

2.2. Cyber-Physical Systems (CPS), Cyber-Physical Production Systems (CPPS), and cyber-physical-Social systems (CPSS)

Cyber-Physical System (CPS) is the collaborating computational entities connected with physical world and its on-going processes [8], firstly introduced in 2006 [9]. CPS is an integrated system between physical and computational (or cyber) systems by the Internet [10], implementing tight conjoining of and coordination between them. Since then, CPS has applied to various areas including manufacturing systems and human aspects.

Cyber-Physical Production System (CPPS) [8] proposed the combination of CPS and manufacturing technology. The paper illustrated the root technologies for the CPPS and proposed future research topics such as cooperative production systems, fusion of real and virtual systems, and human-machine symbiosis. For real case of CPPS in small and mid-size company, heterogeneous database and insufficient data collection can be overcome by real-time and multi-modal data collection and creation of digital twin [11], and proposed the concept of upgraded learning factories based on the digital twin [12]. From the studies, data interoperability and real-time communication were emphasized for smart manufacturing, which was also considered to this study.

Cyber Physical Social Systems (CPSS) is the combination of CPS and social space [13]. According to a paper [14], it regards human as a part of the system, putting humans in the loop. Artificial Societies, Computational Experiments, Parallel Execution (ACP) approach was invented at the Institute of Automation, Chinese Academy of Sciences, for management and control CPPS. Unlike the operation and management of traditional CPSs, humans in CPSSs can be viewed as not only service consumers but as service providers. To my knowledge, there are rare studies of directly combining CPSS to manufacturing, however, much of CPPS works indirectly emphasize the role of human as component of CPS in manufacturing [15].

The motivation of I^{2}CPS framework is how human can be involved to CPPS, therefore it lies between CPPS and CPSS as shown in Fig. 3. The contribution of I^{2}CPS framework on CPS can be described that it proposes combination of middleware and information protocol to solve the issue of data interoperability between physical systems, cyber systems, and human. In addition, I^{2}CPS framework provides VR interface to provide the collaborative space between human and CPS.

2.3. Digital twins

Digital twin is firstly introduced as a fatigue life prediction model and the transfer function of developed stress on aircraft updated by a database of real-time sensor data, thereby estimating the structure life virtually [16]. Therefore, a digital twin is defined as the virtual representation of physical entity evolving through the product life cycle [17]. In manufacturing area, various applications of digital twin have been introduced as showcasing products, process control, and predictive maintenance. The application of digital twins in manufacturing [18–20] shows the common framework skeleton: physical system, virtual system, and model update by real-time information. Physical system generates information consisting of sensor and machine data which is collected in real-time. Virtual system has various types of models: 3D geometry, dynamics, materials, simulation, and analysis. The virtual models respond as the result of diagnosis, optimization, and prediction, then the model is updated utilizing its responses and (or) the real-time information.

The function of I^{2}CPS is for digital twin is that the response of digital twin is overlaid to the VR interface that humans can communicate with autonomy. In the same way, the digital twin can be updated from human data. Another function of I^{2}CPS for digital twins is that the flow of information is divided by its urgency and context. Raw data with high urgency is communicated via M2M middleware while structured information with less urgency uses data-to-information protocol. In summary, digital twins in I^{2}CPS can evolve with human data and structured information.

However, the demonstrated cyber objects such as virtual robot model and conversion model of human trajectory to robot language can be viewed as digital shadow and digital model, respectively [21]. From the definition, digital shadow is a digital object that is automatic updated by physical object and manually update it in opposite. Similarly, digital model has bi-manual update structure. In the demonstration of this study, robots in virtual reality interface are updated from sensor data in real-time. Operators can work with the virtual robot from the real-time established model, and the virtual human works are interpreted to control robot directly or converted to the correspond robot language. Therefore, the terms of digital shadow and digital model were used instead of digital twin to the demonstrated cyber objects in this paper.

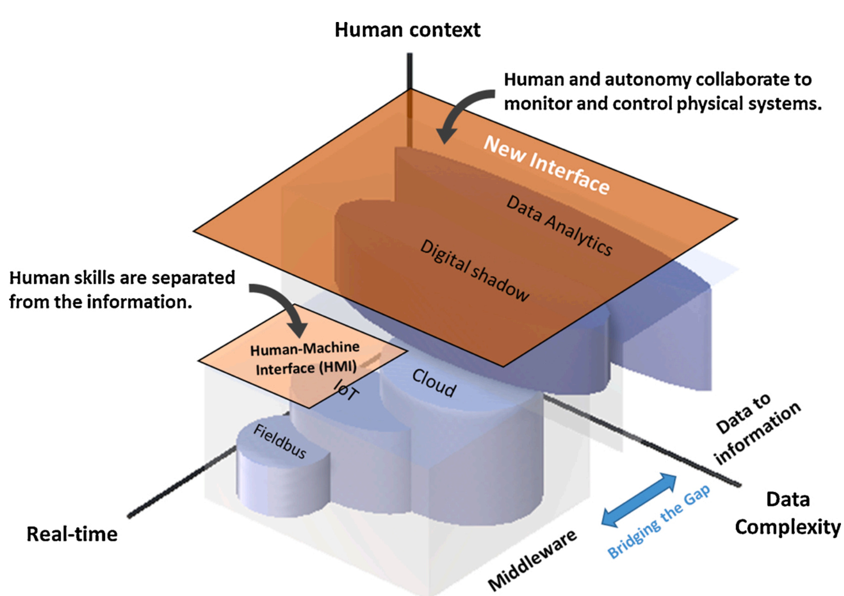


Fig. 2. Data interoperability of manufacturing technologies and interfaces.

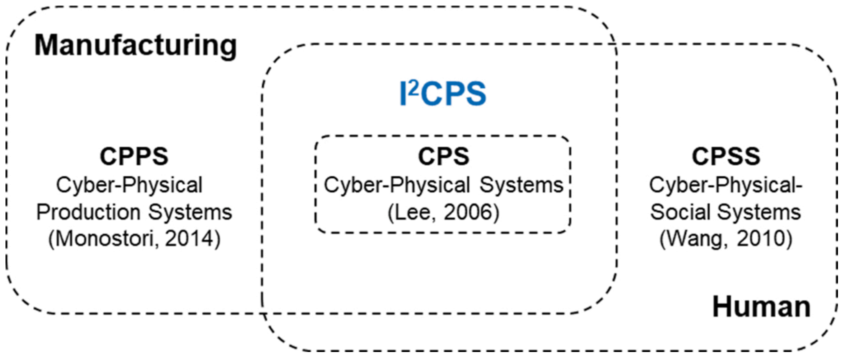


Fig. 3. Diagram of relation among CPS, CPPS, CPSS, and the proposed I^2CPS framework.

Virtual robot objects implemented in this paper were termed as digital shadow and the conversion from virtual human trajectory to robot language was defined as digital model.

2.4. Middleware and data-to-information protocol

In smart manufacturing architectures, data from machines and sensors in physical systems is collected at the same time. It is difficult to construct manufacturing systems with a specific vendor because the complex processes require different types of machine and sensors. Therefore, a bridge is required to connect various communication protocols and to unify data types into a single standardized format, which is one of the main goals in smart manufacturing [22].

The communication between machines is supported by middleware such as OPC UA, ROS, MQTT, and DDS [23]. Basically, they have publishers and subscribers that send and receive data. The connection between them is managed by the main agent (MQTT, ROS) or data itself includes address of the node (DDS, OPC UA). They connect physical assets with different vendors and fieldbus, enabling flexible plug-and-play of manufacturing systems. For example, several studies demonstrated the use of ROS for smart manufacturing. Robot Operating System (ROS) is an open source and well-known framework for cloud robotics [24]. The popularity provides modules of various sensors, robots, and applications, which makes system construction conveniently. For example, robot welding process for the monitoring [25] and simulation for educational purpose [26] were modelled by ROS and game engine (Unity). ROS combined with lightweight virtualization was also utilized to build the CPS of automated guided vehicles (AGVs) [27].

Although the middleware provides such data interoperability, storing and organizing the data for cognition activities are required for advancing human-machine interaction. Previous introductory section addressed the issue that the data structuration is required to cover all the range of data types from raw to structured. Fieldbus, IoT, and cloud help communication among machines, and the data can also reach to human. However, conversion of data to meaningful information for digital twin and other analytic methods is additionally required for smart manufacturing. In addition, raw data also should be converted to human readable format. For example, spindle speed value itself becomes as human context when a unit (rpm) is attached to the value. Such data organization in manufacturing has been performed by several protocols (MTConnect, AutomationML) that converts data into interchangeable format such as XML (Extensible Markup Language) and JSON (JavaScript Object Notation).

Implementing the architecture using MTConnect protocol has been studied recently by numerous researchers. MTConnect provides XML conversion and connectivity tools such as adapter and agent as opensource. Liu et al. [28,29] proposed Cyber-Physical Machine Tools (CPMT), a MTConnect-based platform to utilize the smart manufacturing architecture for a 3-axis milling machine. The concepts of using smart HMIs like VR, AR and mobile devices are illustrated, however, the web-based applications were presented. Another MTConnect-based platform [30] inserted the resource virtualization layer between the physical layer and cyberspace layer to provide web-based service that operators can schedule jobs of robots and 3D printers. To implement the function, additional TCP/IP sockets are added to physical assets because MTConnect is read-only protocol. Álvares et al. [31] combined OPC UA middleware with MTConnect on a CNC lathe to overcome the issue. In summary, MTConnect is useful to collect data from various physical entities and convert to human-readable information, XML documents by the request of applications. However, duplex communication is missing that integration of CPS is limited compared with Machine-to-machine (M2M) middleware.

Blending multiple communication protocols becomes common for edge-to-cloud computing, depending on the location of computing resources [32]. For example, MTConnect has the standard of cutting tools, robots, and sensors and it is also read-only architecture to ensure safety and simplicity of the framework. However, it is not desirable to communicate raw data with each other because MTConnect constructs the raw data to information more as human context. In that case, middleware such as OPC, ROS, DDS, MQTT are suitable. Therefore, combining data-to-information protocol and middleware can construct generalized smart manufacturing platform [32]. Several studies presented the combination of OPC and MTConnect. Álvares et al. [31] combined OPC UA middleware with MTConnect on a CNC lathe machine. Liu et al. [33] developed a Cyber-Physical Machine Tools (CPMS) using OPC UA and MTConnect. Other than the research, this study proposes combination of ROS and MTConnect to communicate wide range of information context from rendering 3D visualization in VR interface and human-robot information context conversion, which is new from other studies.

2.5. Virtual reality-based robot interface

There are 3D interfaces to operate robots both as opensource (RViz MoveIt, Gazebo) and commercial software (RoboDK, Octopuz, Siemens NX). They provide offline simulation, path programming, and deployment with overlaying images and 3D point clouds, which shows more intuitive interface than manual teaching. However, when humans sense the depth of objects, the intuition can be improved. For the reason, virtual reality (VR) has been utilized in this study to enhance the human intuition and interaction with virtual entities. Some studies demonstrated interactive interface for robot manipulation [34,35]. The virtual robot system was rendered after receiving the joint angle from controller. Next, operators manipulate the virtual robot with the visual feedback of current robot posture using intuitive ways such as dragging end effector, choosing 3D pop-up menu, and VR controllers. The robot was manipulated from the virtual command while the movement was visualized again, thereby real robot was operated by feedback and control of virtual robot.

Virtual Reality (VR) was also used for teleoperating robots to gain advantages of immersive 3D environment to users rather than 2D images. Bian et al. [36] introduced the teleoperation of a two-arm collaborate robot with VR headset and human gesture. The VR headset provides 3D vision to a user to provide natural manipulation of the robot. The user can also operate the robot with both hands, and the robot mimics the behavior after extracting the position of hand joints using functions in a Kinect sensor. Similarly, Lipton et al. [37] proposed the VR teleoperation of a two-arm collaborate robot. A game engine (Unity) and VR headset were used for creating effective teleoperation environment. Robot arms, cameras, and sensors were dynamically mapped to user's control room. The system showed advanced manipulation compared with direct teleoperation. Recently, the VR interface was combined with haptic device to provide augmented environment [38]. The studies of teleoperation using VR utilized virtual model for operating robots.

As the utilization of VR model in manufacturing, simulation and education of the systems were suggested. Perez et al. [39] developed VR environment for robot control and operator training. The robot environment was created by game engine, and an operator manipulated robots by interactive user interfaces similar as HMI. Wang et al. [40] introduced a remote welding system using a VR headset. The robot with a welding torch was programmed to move the tool across the trajectory while the operator controls the speed of the torch with camera images and parameters of arc welding overlaid on the VR model. The system separated workers from danger and reduced the inconsistency of width in weaving trajectory. For the design of collaborative workspace, Malik et al. [41] proposed VR assisted framework. There are three steps of the design process. At the ELEMENT-1, product and process models are established, and related resources for robots and automations are gathered to produce a final design. In ELEMENT-2, simulations inside the design are performed while human data generated by RGB-D motion sensor or VR headset are added. In ELEMENT-3, users interact with the simulation to verify the efficiency of human-robot collaboration. However, collaborative robots are the only dynamic model of the virtual entities.

As shown above, there have been efforts using VR to visualize robots or their manufacturing processes intuitively. Based on the previous studies, this study is motivated on how human ability, experience, and knowledge can be maximized and how they can be communicated with physical systems by collaboration with cyber systems. To boost the collaboration between human and cyber physical systems, the framework can visualize not only states of physical assets themselves but also cognitive results from autonomy. In teleoperation mode of robots, for example, the raw joint angle values can be communicated between VR and robot controller because the VR model can draw the robot geometry directly from the values. However, it is possible that degradation of motor and gearbox may be visualized over a robot model after the analysis of joint torques, joint angles, vibration sensors, and other sensor measurements from big data. Another possibility is that virtual human works can be recorded, structured, constrained, and converted to robot works and vice versa, enabling the digitization of human works. To achieve it, combination of M2m middleware and data-to-information protocol is suggested. In summary, this study focuses on proposing the advanced framework for data communication and human interface to implement these possibilities, which is different from previous studies about VR-based robot interfaces.

2.6. Model-Mediated teleoperation (MMT), human-robot collaboration (HRC), and tactile internet

Robot teleoperation consists of a human operator that issues commands, a haptic interface or master device, communication channel, and robot in remote place [42]. It has long history of research to solve the issue of master and slave model uncertainty by disturbance, controller instability by network loss and delay [43], and environment model estimation from external disturbance. One of the recent remedies to guarantee stability and transparency is Model-Mediated Teleoperation (MMT) [44]. In the master and slave side, there are estimation model of counter side that are continuously updated and controlled via the estimation, thereby reducing dependency on the network. The research of MMT has been continued for imitation learning of VR-based teleoperation [45] and human-robot shared control by learning from demonstration [46,47].

Another related research to I^{2}CPS framework is the part of Human-Robot Collaboration (HRC). There are efforts to operate robots with human motions or activities but mostly hand or body positions and gestures. The research mostly utilizes non-contact methods such as image based one (markers, depth sensors, and stereovisions) and wearable one (gloves and bands) [48]. The research mentioned earlier [36–38] are also related to HRC. In addition to measuring positions, classifying types of gestures using machine learning techniques is also promising research to link between human works and robots.

With the advancement of communication such as 5 G and Wifi6, the concept of tactile internet enables real-time monitoring and control of remote robotic systems via mixed reality [49]. It provides low-latency and ultrareliable communication [50] to solve the previous stability and transparency issues caused by network problems. Although few cases are reported for robotics yet [51], it is expected to draw attentions in the future.

The similarities of the above studies and I^{2}CPS are the existence of virtual model, human interface, and robot control by human information. However, the difference of I^{2}CPS from above research area is that it does not focus on real-time or shared control but the issue of human-autonomy-physical system collaboration: issues of communicating heterogeneous information context between CPS entities, human and autonomy collaborate to improve performance of robotic manufacturing, and context conversion between human and robot.

3. Immersive and interactive CPS (I^{2}CPS) framework

3.1. Concept of I^{2}CPS framework

The key concept of I^{2}CPS is the combination of middleware and data-to-information protocol to communicate various data contexts. Fig. 4 illustrates the concept that raw data is communicated via middleware while the data is organized by the separate protocol. The raw data is for fast transfer to the interface such as drawing robot geometries to VR and move the end effector of an industrial robot. On the other hand, the protocol collects data from the middleware using its publisher and subscriber. The protocol is to convert data into the information that both human and autonomy can read. It also stores the data as organized format to provide requested information from the data pool. This concept emphasizes on providing the information with different needs to both human and autonomy enabling seamless collaboration.

3.2. Combination of ROS middleware and MTConnect protocol

For the I^{2}CPS, ROS and MTConnect are used as middleware and data-to-information protocol, respectively. In the I^{2}CPS, machine and sensor data are visualized in the interface while human is reacting in the interface. Commands performed in the interface also should be transferred back to multiple nodes simultaneously. ROS provides tools for robotic applications such as hardware abstraction, low-level device control, message passing, etc. ROS master manages communication between two nodes: publisher and subscriber. The two nodes are decoupled that the publisher does not check if data was received by the subscriber. The communication is not perfectly reliable, however the communication with simple message structure enables fast communications between various nodes. In addition, MTConnect is selected for the standardized manufacturing data protocol. The standard enables developers and users to concentrate on making applications with less concern of device model, brand, and communication method.

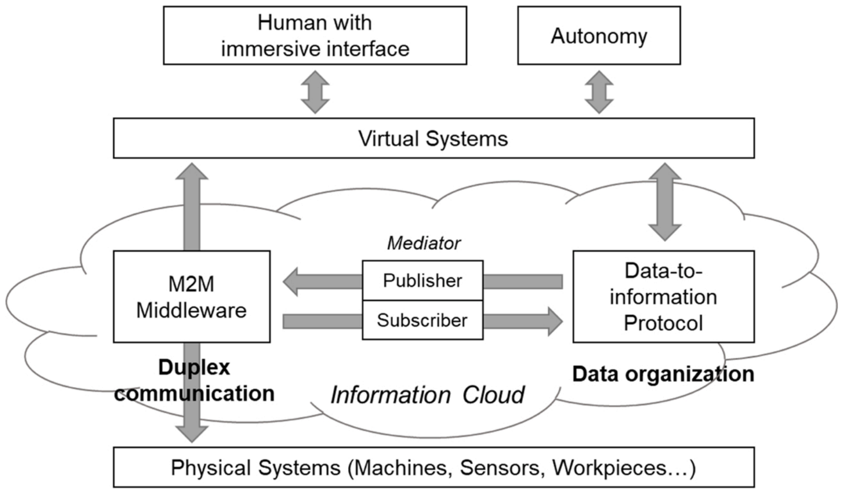


Fig. 4. The concept of I²CPS framework.

MTConnect consists of an agent and multiple adapters for each machine. The agent collects data in memory queue and replies to application's request by XML document format, predefined for the demand of manufacturing applications. The adapter is connected to controllers of manufacturing equipment and the data is collected by the agent. The agent holds the data with its device ID, timestamp, unit, and value range. When the application such as dashboards request the information, MTConnect agent can provide it as XML documents supported by the most programming languages. The XML based MTConnect standard provides structured and contextualized data format from manufacturing equipment. Although MTConnect can receive data from various physical assets, however, it is not suitable for M2M communication. MTConnect works as one-way communication that collecting data is available but receiving data is not implemented. Each physical asset should have a separate data receiving channel for M2M. Furthermore, applications using MTConnect receive XML document. Reading a single data is not efficient because the applications should parse XML headers for each time. The characteristics of ROS and MTConnect are summarized in Fig. 5(a).

Therefore, the link between middleware and protocol uses the publisher and subscriber of ROS and the adapter of MTConnect as shown in Fig. 5(b). It shows that the subscriber of ROS receives the raw data, then MTConnect adapter categorize it before sending it to agent. On the other hand, publisher in the converter requests data to MTConnect agent, receives an XML document, and send the data to any subscribed nodes after parsing the XML.

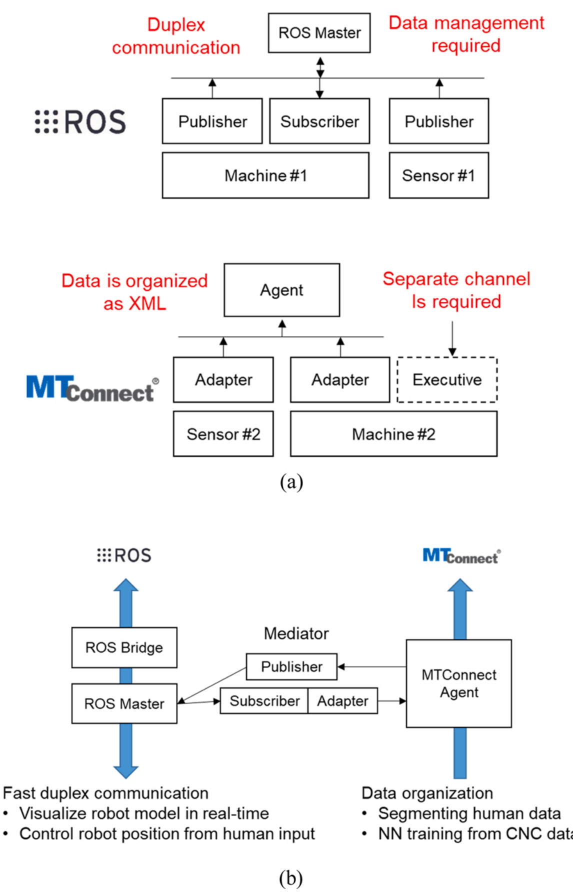


Fig. 5. (a) Advantage and limitation of ROS and MTConnect; (b) Integrated of ROS and MTConnect for I²CPS.

3.3. Integration of I²CPS into smart manufacturing platform

Fig. 6 shows the framework of the I²CPS. The concept of the I²CPS is embedded into a common smart manufacturing platform that mentioned in the previous section. Physical layer includes all the physical assets such as machines, robots, and sensors. The physical assets are connected altogether in the cyberspace gateway layer. Next, cyberspace layer extracts data for decision making and optimization, modeling and simulation, data exchange and integration. The converted information is utilized to various applications by business and consumer side. Other researchers [7,52] also proposed the similar architecture for smart manufacturing: physical layer and cyberspace layer. The difference of I²CPS from general CPS is that immersive interface provides human-cyberspace collaborative space where human and autonomy can communicate.

I²CPS is divided into four parts: 1) smart connection, 2) information cloud, 3) cognition and configuration, 4) collaborative environment. The naming convention is referred from the 5-level CPS structure [21].

First, smart connection of physical components means flexible addition or removal of assets (plug and play), standardized data protocol, and connectivity between physical assets (M2M communication and Tether-free). In I²CPS, each machine or sensor has publisher and(or) subscriber of ROS. In ROS, machines and sensors generate data, while machines also need to subscribe command signals to operate.

Second, data from various physical assets can be retrieved by ROS, creating information cloud. ROS master manages direct communication between publisher and subscriber nodes. Rosbridge [53] is a tool to

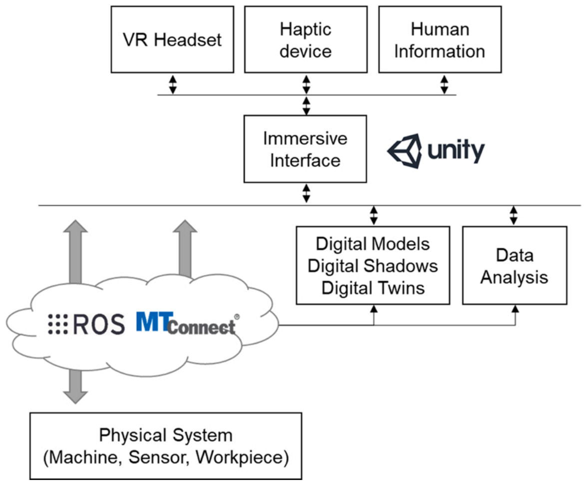


Fig. 6. Framework of I^2CPS for smart manufacturing.

connect ROS master to external network via WebSocket protocol. It is a full-duplex protocol between a web server and a browser based on TCP, therefore any applications connected to Internet including interactive interface, dashboards, and data analysis can use the protocol. The message types in Rosbridge are defined as JSON file format, a message type that has less header information than XML. MTConnect can collect data directly from adapters in physical assets. In this framework, however, MTConnect does not collect data from physical assets but from ROS by the data exchange module as described above. MTConnect is used to convert raw data to XML structured hierarchical data type, providing convenient access to the data. Data required for analysis, prediction, and decision are managed by MTConnect agent, and further long-term data are saved to the database. In summary, the cloud separates data usage into three categories for its timeliness: 1) ROS master handles urgent and single messages; 2) MTConnect agent saves queued data for real-time analysis with the history of data; 3) Database preserves data in the longer term for the prediction and decision making.

Third, virtual entities such as forward and inverse kinematics of robot, work constraints, feedback generation, sensor value interpretation are positioned. Those entities communicate with machines, sensors, human interface, and other virtual entities to provide the required information. The virtual entities are created from programming language and operated as PC application with Ethernet based communication.

Fourth, as collaborative environments, immersive and interactive interface is demonstrated as Virtual Reality (VR) supported 3D visualization of physical assets for manufacturing, consisting of digital shadows and digital models of physical assets. Therefore, users interact with the visualization in virtual environment. The visualization includes geometry of the machine model and overlaying useful information. At the same time, the interface provides an environment that human can operate the machines, work on manufacturing processes, and transfer visual and tactile feedback to user.

Virtual entities access to data from ROS, MTConnect, database, and other applications. From the information, game engines construct the 3D model of physical assets. Game engines are originally intended to develop video game software. However, the applications of game engines are extended to create digital shadows of architecture, product, urban planning, and manufacturing. The game engines have a 3D renderer that developers reduce the burden to write codes to access to CPU and GPU directly for rendering models. Additional advantage of the game engines is that it supports VR environment. Operator can change view angle and interact with 3D objects established in the game engine. In addition to VR devices, other human input devices such as keyboard, mouse, and haptic pen are possible to be integrated to VR. Operator's virtual works can be converted to actual works for physical systems in intuitive ways for humans. Therefore, cognition and configuration are performed from the human-autonomy collaboration.

4. Application of I^2CPS framework

4.1. Rendering robots and generating its intuitive manipulation interface

For the application of I^2CPS framework, industrial robots were rendered in virtual reality (VR) with the same configuration of the physical system. Fig. 7 shows the procedures of how an industrial robot (Stäubli TX2−40) was rendered to the game engine. The corresponding CAD model was decomposed into moving groups and saved as mesh models (STL file format). After the models were imported, they were transformed according to the robot's geometry. Using ROS, six joint angles $q_1 \dots q_6$ were retrieved from the robot controller for every 0.1 s. Next, each component was translated or rotated according to the received joint angle values and the robot's own design information. Due to hierarchical structure of components in Unity game engine, transformation of the former joint affects all the other sub-joints, which represents the serialization of homogeneous transform from the robot base to end effector.

$$\boldsymbol{T}_6^0 = \boldsymbol{T}_1^0 \boldsymbol{T}_2^1 \dots \boldsymbol{T}_6^5$$

$$\boldsymbol{T} = \begin{bmatrix} \boldsymbol{R} & \boldsymbol{t} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}, r_{ij} = f\left(\boldsymbol{q}, \boldsymbol{\eta}\right)$$

where $\boldsymbol{T}$ is homogeneous transform matrix, $\boldsymbol{R}$ is rotation matrix, $\boldsymbol{t}$ is translation vector, **q** is joint angle, and $\boldsymbol{\eta}$ is joint offset. $\boldsymbol{t}$ and $\boldsymbol{\eta}$ are constants which are already defined from the robot's design. The transformation changes the pose of robot's components real-time that VR users can see a synchronized scene with the physical system.

In this study, several types of robots are rendered in VR space. Any robots are possible to be rendered since it provides kinematics and mesh-based 3D model. Most robot vendors provide CAD model, and the CSG (Constructive solid geometry) models can be converted to mesh models by various software. In addition, URDF (Unified Robot Description Format) information of robots is also shared in public that kinematics of robots can be imported. After 3D entities are positioned, translational or rotary joints can be defined by joining a C# script with joint definitions for each parent game object. Fig. 8 illustrates various robot models constructed in the VR space showing the flexibility of adding robots to the interface.

Next, controlling the pose of the end effector in the I^2CPS was implemented. In the framework, a colored box was movable by a VR controller and robot's end effector follows the box's position and orientation. It was implemented by inverse kinematics calculating six joint angles from the pose of the box. Inverse kinematics for robot with spherical joints, usually common in industrial robots, can be solved by geometric method [54] from the fact that XYZ position after joint #1,2,3 and spherical joint from joint #4,5,6 are in same location.

$$\left(\boldsymbol{T}_1^0 \boldsymbol{T}_2^1 \boldsymbol{T}_3^2\right)^{-1} \boldsymbol{T}_6^0 = \boldsymbol{T}_3^4 \boldsymbol{T}_4^5 \boldsymbol{T}_5^6$$

where $\boldsymbol{T}_6^0$ represents the colored box's pose and the remaining calculation is to compare elements in both matrices to solve six joint angles ($q_1 \dots q_6$). One thing to note is that there are eight possible solutions after calculation, and a unique solution should be selected by providing the appropriate solution space.

Fig. 9 illustrates how the pose control function was utilized in the VR interface. A white line with a round end was changed by VR controller. When the white dot indicates the box, humans dragged the box while pressing a trigger button. In "Controller mode", the movement of the colored box was restricted except predefined functions like moving the robot (M0609, Doosan robotics) to measure objects from a Lidar sensor. However, it is possible to move the robot in real-time by converting the

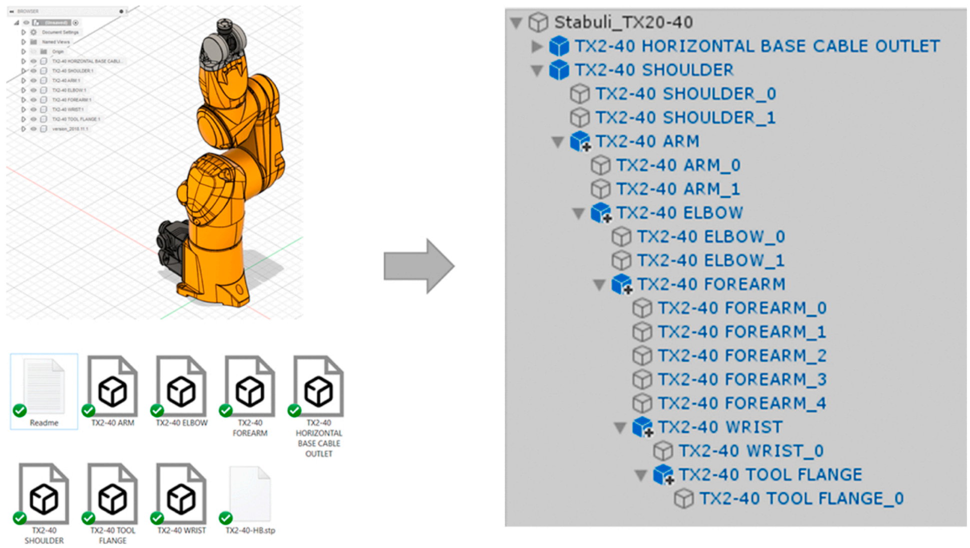


Fig. 7. Generation of geometric robot model in Unity.

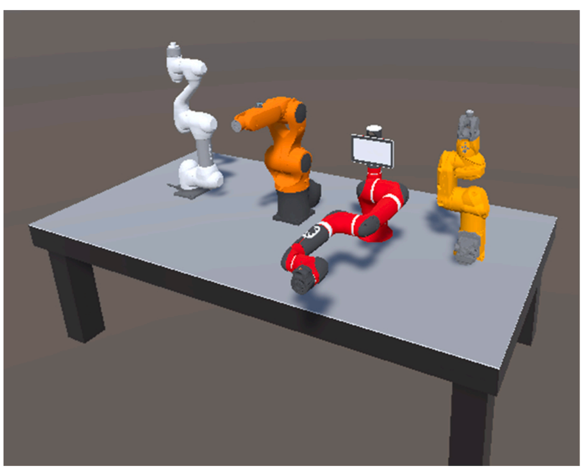


Fig. 8. Various robot model in VR interface: (From left) Doosan robotics M0609, Kuka R6 700, Sawyer robot, and Stäubli TX-40.

box pose to joint angles by inverse kinematics, then send the joint angles to the robot controller via ROS. The refresh rate was restricted by 0.5 s due to calculation time for inverse kinematics and internal logics of robot controllers. In "Simulator mode", the robot follows the colored box freely without moving the robot, and the refresh rate was 0.1 s.

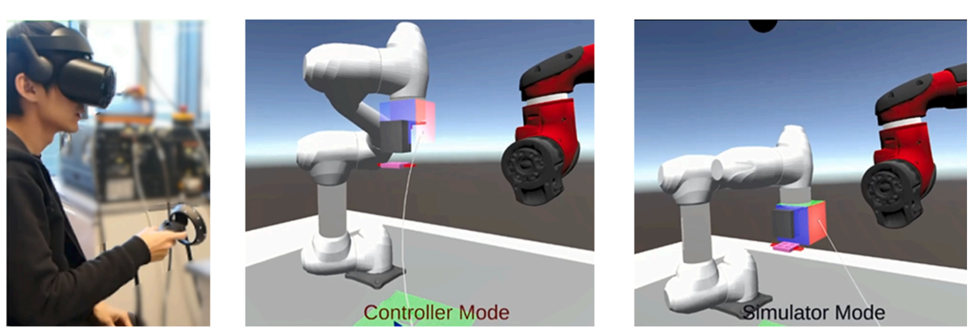
4.2. Constraints of virtual human works

In the I^{2}CPS, humans work virtually in the VR interface. Since it is a virtual space, trace of human works will be mapped to physical space. More than direct one-to-one mapping, various visual aids, constraints, and variable space mapping in virtual space will improve the accuracy of human work and prevent collisions. With the objectives, this section demonstrates augmentation of human works via the interface.

Visualization of information on the VR interface can help humans to work inside the virtual space. One of the objectives is to illustrate the movement range of robot arms. It can prevent alarm stops when joint angles outside the given range are sent to the controller. In addition, collisions can be prevented if the space is well defined to avoid obstacles. In this application, the working volume of a robot arm will be rendered as a closed mesh model. It will be overlaid onto the VR interface, and collision check between robot end effector point and the defined volume will be performed so that position input to the controller outside the volume is prevented.

As a result, maximum points of the end effector were drawn, then the points were used to create triangular meshes. Fig. 10 illustrates the information about maximum and minimum robot range in XZ plane. Based on the information, the maximum range was calculated, and the corresponding vertices were generated. Next, the planes were rotated with Y axis which corresponds to the joint #1′s angle range. At that time, as in Fig. 11, the vertices on the planes were copied to create a point cloud.

From the 3D vertex cloud, triangular meshes were generated as in Fig. 12. The created vertices were ordered and grouped into each tringle that the normal of the mesh points outward the volume. The meshes



(a) (b) (c)

Fig. 9. (a) Human operating robot in I^{2}CPS; (b) Moving a robot is restricted in controller mode; (c) The robot can be movable freely in simulator mode.

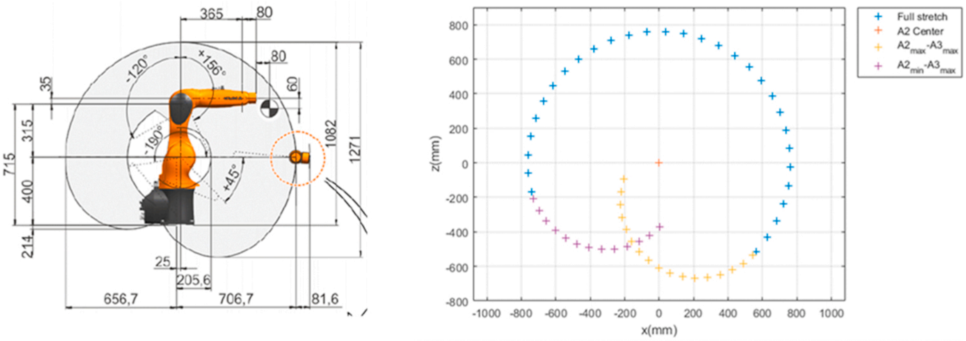


Fig. 10. The maximum range of robot in XZ plane and the generated vertices.

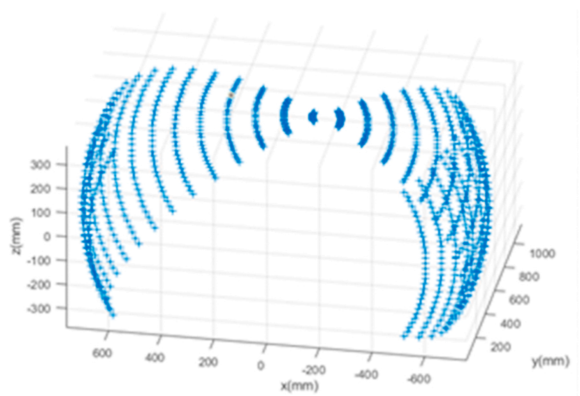


Fig. 11. A point cloud after rotating the points with the 1 st joint angle range.

were generated until a closed volume was created. The whole procedure is implemented as C# script in Unity that the volume was automatically generated for selected ranges of joint angles.

Another example of visualizing machine data is to indicate joint torques during robot movement. The color was set from green to red depending on the amount of joint torque value. Fig. 13 shows that colors of joint #1, #2, and #3 were changed when the joint #1 was rotated. The visualization is intuitive in that human can recognize where and how much the joint torque was generated rather than reading raw numbers.

Applying constraints in VR workspace is expected to improve the accuracy of human work. For example, if input device provides haptic force when the pen touches the mesh surface, it will be used to constrain position of human inputs to follow the surface. In this application, a haptic pen (Touch, 3Dsystems) was implemented inside the VR interface. Fig. 14 shows a schematic diagram how the VR interface communicates pen positions and generates feedback force. There is haptic pen's internal algorithm to generate the feedback force when the defined mesh collides with the points of haptic pen. The refresh rate of the algorithm is 1 kHz, and developers can change the parameters of feedback force such as direction vector, amount, spring constant, and damping constant.

In the experiment, the working environment to touch the surface of slot model was implemented. Next, trajectory of pen input while scratching the surface was recorded via the I^2CPS framework. The demonstration was targeted for robot assisted surface finishing processes. While rubbing the surface, feedback force was generated that the pen cannot penetrate the surface of the slot. Fig. 15. shows the gathered trajectory points and its projection to a plane, respectively. It shows that operator can follow the defined surface at given feedback force constraint. When developing applications of I^2CPS, such constraints in virtual interface will guide human works with improved precision.

In addition to the constraints, variable space mapping was demonstrated. The mapping from input devices to the physical systems is to transform the positions without loss of generality or functional intent [55]. In the VR interface, local input devices have various shape and degrees, and their structures are dissimilar to machines in the factory. For example, Cartesian space mapping from an input device to a 3-axis machine is enough to transfer human movements. However, 5-axis machining center requires to transfer orientation of the end as well as Cartesian positions. Industrial robots have various degree of freedom and arm lengths, which does not match to input devices at the local side. Therefore, space mapping from input device to machine is considered in this application.

Dubey et al. [56] introduced variable velocity mapping that a linear scaling function was parameterized and the coefficients were estimated from several parameter value sets. Chen et al. [57] proposed a hybrid method of one-to-one joint space mapping and operating space mapping of an industrial robot. At the joint mapping stage, devices in master and slave side were linearly mapped to cover the area as possible. When the user requires precise control, operating space mapping method can be

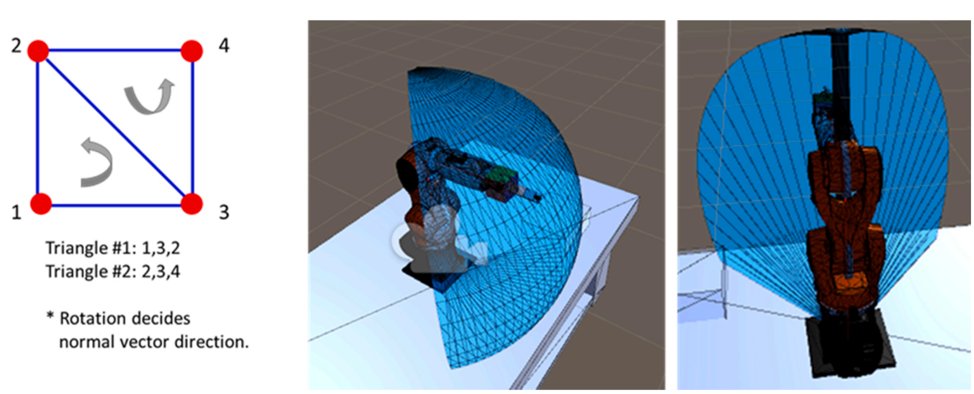


Fig. 12. Mesh creation method and the resulted volume.

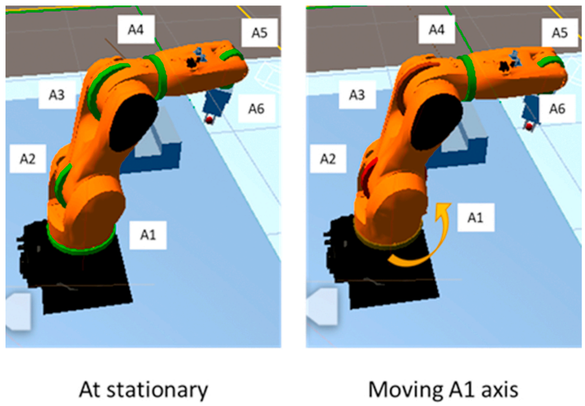


Fig. 13. Comparison of joint ring colors when the robot is stationary and joint #1 rotates.

used instead. In this method, master's joint angles were converted to Cartesian space by forward kinematics, then the space was transferred to the slave side. Next, joint angles of the slave side were calculated by closed-loop inverse kinematics [54] as in below:

$$\dot{\boldsymbol{q}}_s^d = \gamma \cdot \boldsymbol{J}^T\left(\boldsymbol{q}_s^d\right) \cdot \left(\boldsymbol{x}_s^r - \boldsymbol{x}_s^d\right)$$

where $\dot{\boldsymbol{q}}_s^d$ is a vector of joint angles, $\boldsymbol{J}^T\left(\boldsymbol{q}_s^d\right)$ is a pseudoinverse of Jacobian, γ is a constant, $\boldsymbol{x}_s^r$ is desired end effector position from master device, and $\boldsymbol{x}_s^d$ is calculated end effector position of the slave robot using forward kinematics of $\boldsymbol{q}_s^d$, which can be obtained by integrating $\dot{\boldsymbol{q}}_s^d$. Rendered haptic feedback force is also changed when the mapping method is changed. Although the proposed method could achieve wider operation coverage and tracking capability, the space mapping is linear for all the workspaces. The desired position of the slave system is

$$\boldsymbol{x}_s^r(t) = \boldsymbol{x}_s\left(t_p\right) + k_s\left(\boldsymbol{x}_m(t) - \boldsymbol{x}_m\left(t_p\right)\right)$$

where k_s is a constant scaling factor and t_p is previous calculation time instant.

The objective is to develop variable space and feedback force mapping considering manufacturing workers using the VR interface. For example, when the workers try to pick up a part using robot, space on the master side can be enlarged for precise control. After grabbing a part, they wish to move the robot arm quickly to another place, in which sensitivity should be higher than the former case. Therefore, desired position of the slave system $\boldsymbol{x}_s^r$ is modified as

$$\boldsymbol{x}_s^r(t) = \boldsymbol{x}_s\left(t_p\right) + f(\mathrm{x}, \mathrm{y}, \mathrm{z})\left(\boldsymbol{x}_m(t) - \boldsymbol{x}_m\left(t_p\right)\right)$$

where $f(x, y, z)$ is a function of variable scale ratio according to position of master device.

In the I[2]CPS, the variable velocity mapping was implemented as in Fig. 16. In the defined boundary, distance of haptic pen movement is halved that operator feel the pen is moving slowly and precise positioning is possible. Outside the boundary, the distance is not scaled. Such space setup is expected for operators to improve accuracy in the VR interface.

4.3. Deploying virtual work to physical system

The I[2]CPS framework is designed to deploy virtual works from human to the physical systems. To demonstrate the application, manipulating robots in the VR interface was converted to automatic robot program. In the demonstration, a 6-axis industrial robot (KUKA KR6 R 700) and VR headset (Samsung Odyssey+) were utilized. Fig. 17 illustrates the schematic of the application. ROS transfers current joint angles to render the virtual robot model in real time (10 Hz) and desired joint angles to control position of the robot by human input (5 Hz). Joint angles were converted from pose vector by geometric method of inverse

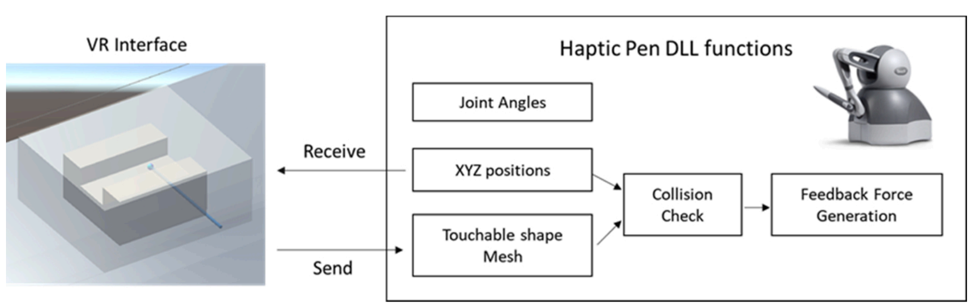


Fig. 14. Schematic of communicating positions and feedback force generation.

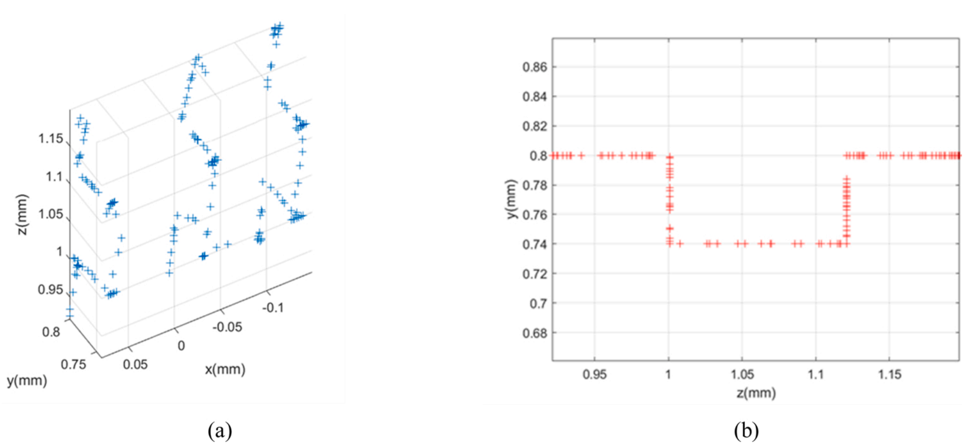


Fig. 15. Result of scratching the mesh surface in VR interface: (a) Recorded points; (b) 2D projection view.

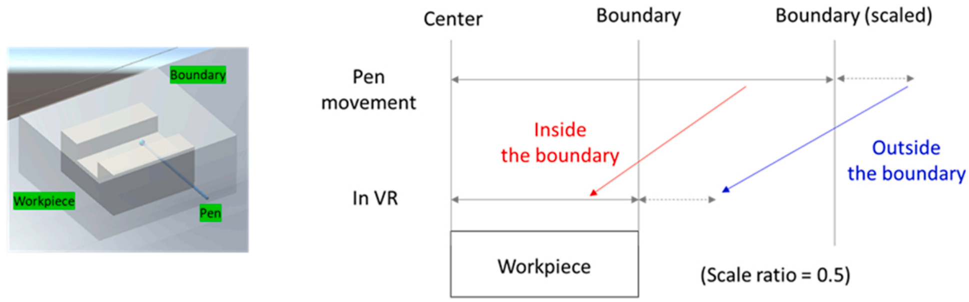


Fig. 16. Variable velocity mapping in I[2]CPS framework.

kinematics [54] with C# scripts in Unity game engine. The robot model in Unity and a PC connected to the robot controller have a set of ROS publisher and subscriber to communicate the raw joint angles. At the same time, the data in ROS was captured by MTConnect adapters. ROS subscribers in MTConnect received the raw data and the adapters converted it to XML language. It categorizes the raw data into device ID and value types, then adds timestamp. Such information structure enables both human and autonomy to find the required data quickly and to understand the data as their own semantics.

In the VR interface, position of end effector in the robot was controlled by translating and rotating a manipulation box as in Fig. 18. By the function implemented in I[2]CPS framework, the robot end effector follows the position and orientation of the box. The position command was sent for 2 Hz because the robot controller's internal interpolation cannot follow the desired position quickly after joint angles are written. The trajectory of the manipulating box was collected by MTConnect, then the points were saved as an XML document shown in Fig. 19(a). Next, they were parsed to extract the recorded trajectory. After transforming points from VR interface to the robot coordinate, the robot language was automatically generated by converter program made of Python and Matlab script. Fig. 19(b) shows the generated robot programming language, KRL 8.0 standard for the robot. The generated text file of the robot program was either stored in PC and moved manually via USB drive or can be sent to controller PC via Ethernet.

The generator creates spline trajectory based on the collected points, while the splines are separated by duration between points. MTConnect gathered data with the rate of 5 Hz (200 ms), however the collection stops when the value of data item is not changed. The long duration around 2.2 s inside the trajectory data was analyzed as hesitation of human input. Therefore, SLIN command was inserted to create the end of spline and start a new one. Fig. 20 compares the trajectory of human input with the one of robot movement when the generated robot

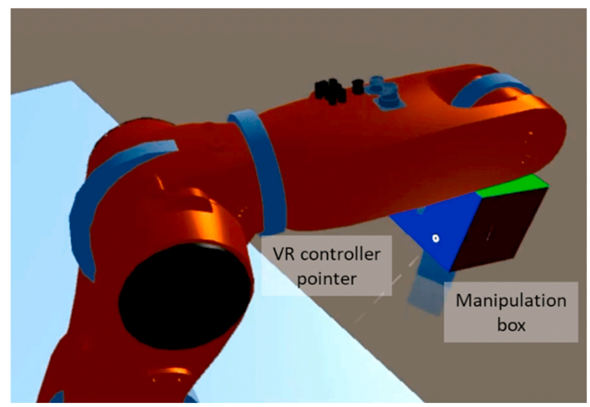


Fig. 18. Illustration of manipulation box translated or rotated by dragging and dropping of VR controller.

program was running. The spline trajectory of the robot mostly follows human inputs. However, the approximation generates difference of 3.64 mm at point A and 4.4 mm at point B. It verifies that human input can be converted to the information for robot, and human work can be automated via I[2]CPS framework

As an advanced application, a collaborative robot (M0609, Doosan robotics) with a pneumatic gripper was utilized to demonstrate virtual pick-and-place teaching and its deployment to the physical systems. In the VR interface, measured point cloud from Lidar sensor (L515, Intel) was transformed and rendered. The user moved the pose of a colored box attached at the end effector to the desired point and pressed the button of VR controller to record the point. Fig. 21(a) illustrates the procedure that constrained roll and pitch of the colored box enables quick and

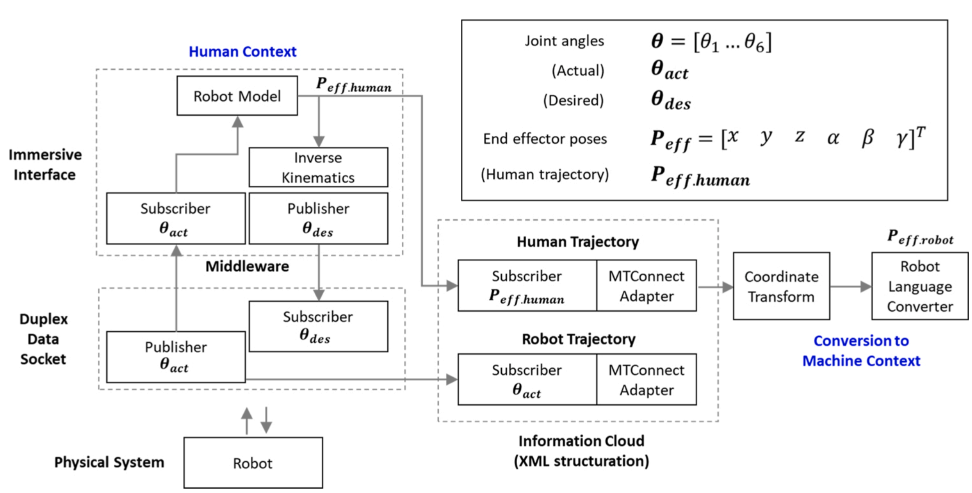
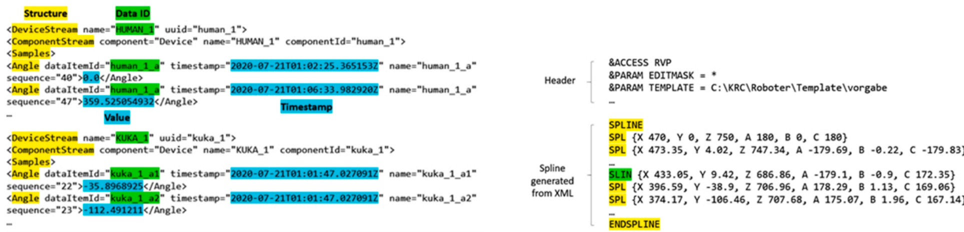


Fig. 17. Schematic of deploying virtual human works to physical systems.



(a) (b)

Fig. 19. (a) XML information collected by MTConnect; (b) Example of robot program language generated from human input trajectory.

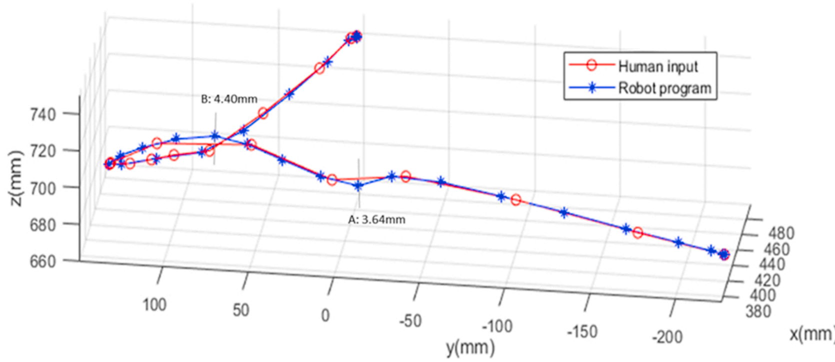


Fig. 20. Comparison between collected human input and robot movement by the robot language.

intuitive teaching for the given pick-and-place setting, which is not provided by direct teaching method of the collaborative robot. Next, the recorded points were converted to the corresponding robot language (DRL language syntax). As in Fig. 21(b), the robot program was executed that virtual teaching was deployed to the physical system.

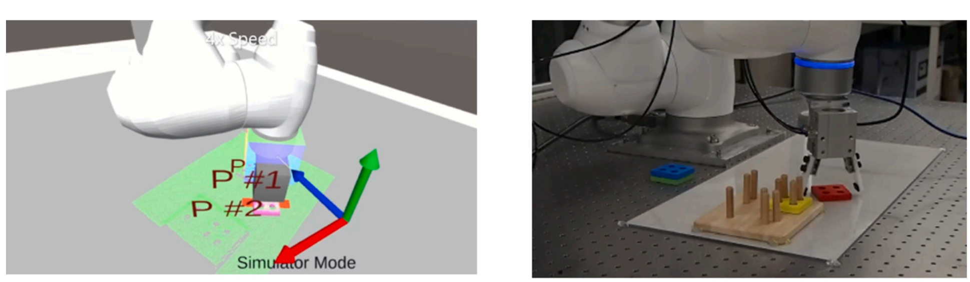
5. Limitations and research opportunities

This study utilized semantic vocabularies of MTConnect, however, it did not focus on information structure of manufacturing in both humans and physical systems. The research progressed from construct manufacturing processes utilizing ontology [58,59], then communication of the information has been expanded from software applications to human side such as operator, maintenance person, designers, process planners, service providers, and consumers [60,61]. The semantics of manufacturing information are implemented for providing intelligent knowledge base. For example, job-knowledge ontology bridged the gap between tasks required in companies and knowledge from vocational education and training [62]. In another research, PSP Ontology (Problem, Solution, Problem-Solver Ontology) was proposed and integrated into CPPS, thereby demonstrated human competences-CPPS autonomy collaboration for intelligent maintenance [63].

Currently, the proposed I^{2}CPS framework utilizes structured information that data-to-information protocols provide. For example, MTConnect standardizes manufacturing machine and process data as semantic vocabularies predefined by XML headers and attributes. OPC provides less constrained XML format to represent the information. The next level of information semantics such as the ontology-based CPS is out of range in this study, however, the I^{2}CPS framework suggested that the protocol can be redesigned to support like the ontology-based information, which is communicated within CPS entities.

In I^{2}CPS framework, it is required to customize the information protocol for given application. As mentioned earlier, new semantics other than given ones from prevailing protocols will strengthen the framework. For example, Job-Know ontology [62] might be useful to construct task virtually from human intents. The Job-Know ontology is useful to describe a job consisting of multiple tasks and know-how of humans, thereby connecting two heterogeneous information. Table 1 illustrates the possible robotic manufacturing applications of I^{2}CPS framework based on the Job-Know ontology. It is expected to transfer human skills and intuitions to manufacturing applications via the framework. There are other research opportunities about how to customize data-to-information protocol to support ontology models and create autonomy to implement such functionalities.

To make I^{2}CPS framework applicable to shop floors, increasing complexity and non-linearity during scaling up the framework should be



(a) (b)

Fig. 21. (a) Virtual teaching in I^{2}CPS interface; (b) Deployment to physical system.

Table 1
List of possible I^{2}CPS application based on Job-Know ontology.

Human Knowledge	Virtual Robot Tasks
When picking up objects on the ground, top shoulder posture of robots is intuitive to human.	Robot is constrained in top shoulder posture when picking up objects virtually.
When teaching robots to pick and place application, it is convenient to hold X and Y rotation and move Z axis.	VR interface constrains end effector's XY rotation and allow to move Z axis when teaching robot pick-and place.
Process designers decide orientation of robotic welding tool for certain geometries based on welding process theory and experimental learning from resultant qualities.	Suggested tool orientations are visualized for selected geometry of target welding points.
Operators decide tool orientation and shape after checking direction and amounts of burrs.	Deburring tool shapes and robot trajectories are suggested and overlaid to VR interface for given burr formation information.

handled. Products, manufacturing processes, control systems, consumers are main sources of the complexity [64]. The I^{2}CPS utilizes manufacturing middleware and data-to-information protocol to unify communication methods, thereby reducing complexity of information network. In addition, object-oriented programming of VR interface enables the concept of plug-and-play, model abstraction, code reuse, and resource management. However, other complexities from products, processes, and shop floors are raised and listed in below.

- Products: Product Data Management (PDM) and Product Lifecycle Management (PLM) are widely used in enterprise to manage complexity of products. This study did not mention the interoperability between I^{2}CPS framework and the management systems. Although adapters between data-to-information protocol and PDM/PLM have been developed, utilizing the PDM and PLM information into the framework is challenging topic. For example, the method to digitize human works and convert to the management software and the opposite way requires new definition of information protocol and creation of adapter software.
- Processes. This paper mainly demonstrated robotic application possible for assembly, pick and place, and finishing operations. However, there is limitation that other processes such as additive manufacturing, subtractive machining, and logistics are not proposed. Although the information of hardware, control software and algorithm, sensors are communicated as a unified way via I^{2}CPS framework and customized per application, virtualizing a large-scale process consisting of multiple sub-processes, products, and human knowledge is challenging for given computing power, network bandwidth, and basic information semantics. Utilizing distributed system and modulization of sub-components might solve the complexity.
- Shop floors. I^{2}CPS framework visualizes physical system and humans work virtually in VR interface. Therefore, maintaining model fidelity is key factor for effectiveness of the framework. Unfortunately, shop floors are changing dynamically by collision, wearing, and production plans, therefore difficulty of the model fidelity increases as physical systems become more complex. Therefore, utilizing sensors and developing methods to find model parameters such as quick machine calibration and of quality information analysis should be implemented further to relieve the complexity.

In this study, 3D models of robots and objects are called as digital shadow because they are constructed from raw data continuously and do not demonstrate two-way and self-evolving characteristics that common digital twins have. The interface can be improved if more digital shadows are implemented in addition to robot posture, allowable workspace at given joint limits, and joint torques shown earlier. For robots, candidate digital shadows are static calibration parameters, bending amounts for attached weights, applied force at end effector tool, the prediction result of preventive maintenance, and so on. Objects and environments also can be digital shadows from sensor data of Lidar, structured light sensors, 2D images, contact sensors, and human inputs. Such digital shadows are expected to improve fidelity of virtual model and assist decision making of humans in the virtual interface.

6. Conclusion

In this paper, an immersive and interactive cyber-physical system (I^{2}CPS) framework was proposed. The objective of the framework is to promote human involvement to smart manufacturing: utilizing strength of human skills like fast and flexible cognition, self-adoption, and self-decision while overcoming limitations of low accuracy, human errors, and big data analysis. To cover the wide range of data context and temporal requirements, middleware (ROS) and data-to-information protocol (MTConnect) were combined in the framework. In addition, the virtual reality (VR) interface was implemented for collaboration of human-machine-autonomy. Several applications were demonstrated to verify the effectiveness of the framework. For constraints of human works, setting robot boundary, visualization of robot joint torque, utilization of haptic feedback, and variable velocity mapping were demonstrated. Such constraints maximize or restrict human works for improving accuracy and reducing human errors. Other applications demonstrated how the virtual works were deployed to physical system. Human trajectories were collected and converted to autonomous robot program that autonomous entities are generated from human skills.

Future works of the I^{2}CPS framework are enumerated as in below.

- New middleware and data-to-information protocols will be studied to construct more effective framework.
- Complex and manual manufacturing processes will be implemented by the framework. For example, finishing processes (deburring, polishing, grinding) relies on human skills for parts with complex geometries. Robot assisted autonomous system with I^{2}CPS is expected to provide opportunities to digitalize human skills. Combination of the digitalized autonomy and human skills will improve process productivity and quality together.
- Improving model fidelity between the VR interface and the physical systems will be studied. For example, robots in VR interface are rendered with designed geometry values. However, actual robots have errors from the model due to inaccurate assembly and tolerance of parts. Therefore, accurate and affordable calibration methods will be studied for precise deployment of virtual works to the physical systems.

Data availability

Data will be made available upon request.
All data is within the manuscript and figure.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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